

Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

The First Computers...



Learning Outcome:

Identify information about the history of software and its characteristics in simple spoken and written situations.
Give information about software history, characteristics and methodologies.

Ask your partner:

- What were the most common characteristics of the first real computers?
- When did the mouse and graphical feature in a computer appear?

Can Do Statements:

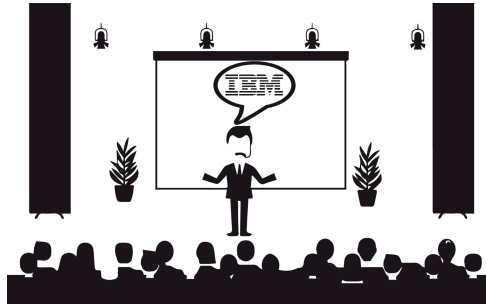
- I can understand written information about a successful family of computers.
- I can use vocabulary related to mainframe architecture.
- I can write a simple text about the importance of mainframe architecture.





Let's Explore!

The Beginning of the Modern Computer Industry



Instructions: Read the text below.

FORTRAN ("FORmula TRANslator"), the world's first actual computer language was released in 1954 and, thanks to John Backus's vision, its IBM developer, it allowed people to program computers with simple instructions similar to the English language and with mathematical formulas. FORTRAN's popularity led to the creation of COBOL (Common Business-Oriented Language) and together, they dominated programming until the end of the 70s.

Although these two early programming languages improved programming functions, there was not an actual operating system as we know it today. Everyone agrees that MAD (Michigan Algorithmic Decoder), developed at the University of Michigan in 1959, is the first true operating system. However, the concept of the operating system wasn't very well known, until the influential development of IBM's S/360; The IBM System/360 and OS/360 mainframe architecture mark the beginning of the modern computer industry.

On April 1964, IBM launched the System/360, this model and the next versions of it, were the first successful family of computers ever. These 19 models had a consistent architecture in computers of different sizes. Their features were the use of the same peripherals and software. Therefore, it was easy to move onto new models as they were released. The idea was revolutionary because, before that, S/360 users had to rewrite their entire applications and replace the peripherals.

The IBM System/360 mainframe architecture is a set of distinct terms and rules that are instructions to build products. In computer science, an architecture defines the structure of the computer system. Each generation of mainframe computers includes improvements in architecture to remain stable, secure, and compatible with as many computing platforms as possible.

Some people say that the S/360 IBM became the most successful computer in history thanks to its mainframe architecture. The importance of this architecture is that it presented a number of advances, such as the first transaction processing system and the first use of solid logic technology (SLT). However, these computers were not error-free and problems were not so easy to solve, the peripherals performed poorly, their processor was slow and their communication capabilities were virtually nonexistent. In spite of all of this, this mainframe architecture introduced the world's first sophisticated operating system. For the first time, compatibility was possible and a new part of the computers history began. With the S/360 and OS/360, the operating system is now the key unique factor of a computer system.



Let's Practice!

IBM: A Company of Great Family of Computers

Instructions: Write the words to correctly complete the sentences. Use words from the box below.

creates / establish / have

1. The IBM System/360 and OS/360 mainframe architectures _____ the beginning of the modern computer industry.
2. In 1964 IBM _____ the System/360.
3. System/360 and OS/360 _____ a consistent architecture.



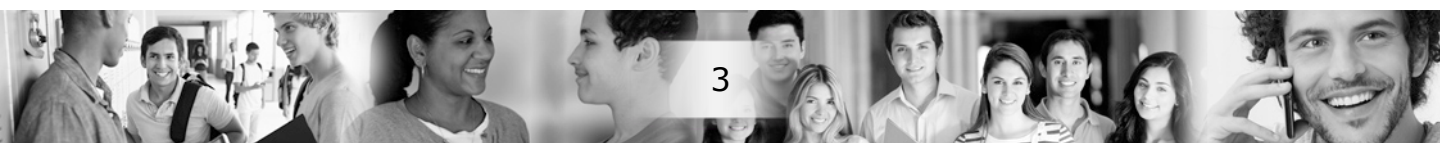
Let's Practice!

Mainframe in Computer Science

Instructions: Read the following sentences based on the text. Circle True or False.

The importance of mainframe architecture is that it presents a number of advances, such as the first transaction processing system, the first use of solid logic technology (SLT) and represents the world's first sophisticated operating system. However, these computers were not error-free even when problems were not difficult to solve, the peripherals performed poorly, their processor was slow and their communication capabilities were almost nonexistent.

1. The first transaction processing systems weren't available in the mainframe architecture. **True False**
2. The first use of SLT was in the mainframe architecture. **True False**
3. The mainframe architectures in the first family of computers were error-free. **True False**
4. The peripherals' performance was not very good. **True False**





Extension Activity

Family Computers Timeline.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for further information on the history of mainframe computers.
- Design a timeline that starts in 1950 and finishes in 2017.
- Put the mainframe computers in the years they were launched.
- Show and explain the timeline to your partners.

Self-Assessment

Yes

No

- I identify when the principal mainframe computers appeared.
- I design a timeline with relevant information related to mainframe computers.
- I carry out a brief presentation on the mainframe computers timeline.

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Target Words

- distinct** (adjective): different from others
e.g. The IBM System/360 mainframe architecture is a set of distinct terms and rules that are instructions to build products.
- remain** (verb): to continue to be the same
e.g. Each generation of mainframe computers includes improvements in architecture to remain stable, secure, and compatible with as many computing platforms as possible.
- mainframe** (noun): a very big computer, regularly the center of a system serving many users
e.g. The IBM System/360 and OS/360 mainframe architecture mark the beginning of the modern computer industry.

- launch** (verb): to release, publish, or send forth
e.g. On April 1964, IBM launched the System/360, this model and the next versions of it, were the first successful family of computers ever.
- release** (verb): to issue, publish, launch, broadcast or exhibit
e.g. FORTRAN ("FORMula TRANslator"), the world's first actual computer language was released in 1954 and, thanks to John Backus's vision, its IBM developer, it allowed people to program computers with simple instructions similar to the English language and with mathematical formulas.



Learning Points

Simple Past Tense of Verb Be

The verb **to be** is normally used to indicate the role or state of a person, to talk about characteristics when it goes with an adjective and to indicate location. This verb has the same uses in its **past form**.

Unlike other verbs in the **past form**, to make the negative form with to be you only have to add the word not after it, as it works both as a verb and an auxiliary:

That **was not/wasn't** the fastest computer back in those days.

Notice how the contraction is made, all you need to do is to put the two words together (**was + not**) and replace the letter "O" in not by an apostrophe.

Take a look at the following chart to see the **negative past form** of the verb **to be** with all the subjects:

CHART 1

VERB TO BE – PAST NEGATIVE FORM		
SUBJECT	VERB TO BE	EXAMPLE
You, We, They	were not (weren't)	<i>They weren't the best computers.</i>
I, He, She, It	was not (wasn't)	<i>I wasn't present when talked about that.</i>

Examples

- *They were not equal in size.*
- *Families of computers were not common in the 1960's.*
- *These computers were not error free.*
- *Problems were not difficult to solve.*





Learning Points

Expression Abstraction in FORTRAN

FORTRAN (word derived from Formula Translation) is an all-purpose, imperative programming language which is used especially in numeric computation and scientific computing and has had a very big influence on computing and programming from the very beginning.

During the initial days of computing, programming was a very slow and discouraging activity that required every single step to be coded as a separate machine instruction. By 1953, seeing this situation, John W. Backus started the development of the **Fortran** programming language (the name being a contraction of FORMula TRANslation).

His team's objective was to create a compiler capable of producing efficient object code similar hand-written assembly code and **Fortran** was exactly what they were looking for. Programmers did not have to write hard-to-understand hieroglyphics when, for example, creating an instruction to divide two variables A and B and saving the result in C, they could now a more natural and easy-to-understand statement like:

$$C = A/B$$

This is what we know as **expression abstraction**, because mathematical expressions could be written very similarly to the way they appear in a textbook. The secret for **Fortran's** initial quick spread was that scientists were able to write programs to solve problems themselves, in a simple way and with limited access to professional programming knowledge. Another key aspect for its success was that a program written in **Fortran** could be moved to any computer with a **Fortran** compiler.

